



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester MCA (CBCS) Degree Examinations

Summer Semester July 2024

22MCA204 – DATA WAREHOUSING AND DATA MINING

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions

Unit – I

	Marks	BT*	CO*	PO*
1. a) Describe fact constellation with an example.	5	L*2	1	1
b) Compare OLTP and OLAP.	5	L2	1	1
c) What is a Data Warehouse? Explain the key features of Data Warehouse. Also explain how organizations use information from data warehouses.	10	L2	1	1
2. a) Describe the ETL process.	4	L2	1	1
b) Explain different categories of measures.	6	L2	1	1
c) Explain major OLAP operations on multidimensional data with suitable examples.	10	L2	1	1

Unit – II

3. a) Explain how OLAP queries can be efficiently processed.	5	L2	2	1
b) What is an attribute? Compare following types of attributes. i. Interval and Ratio attribute ii. Qualitative and quantitative attributes	5	L2	2	1
c) Describe following types of preprocessing techniques. i. Sampling ii. Discretization	10	L2	2	1
4. a) What are datasets? Explain the characteristics of data sets.	4	L2	2	1
b) What is an Aggregation? What are the motivations for this operation?	4	L2	2	1
c) What is data mining? Explain the challenges that motivated the development of data mining.	6	L2	2	1
d) Describe various OLAP server architectures.	6	L2	2	1

Unit – III

5. a)	What is Association analysis? Brief the terms: Itemset and Support Count.	4	L2	3	1												
b)	Consider the following transaction data set.																
	<table> <tr> <th>TID</th><th>itemset</th></tr> <tr> <td>1</td><td>{f, c, a, m, p}</td></tr> <tr> <td>2</td><td>{f, c, a, b, m}</td></tr> <tr> <td>3</td><td>{f, b}</td></tr> <tr> <td>4</td><td>{c, p, b}</td></tr> <tr> <td>5</td><td>{f, c, a, m, p}</td></tr> </table>	TID	itemset	1	{f, c, a, m, p}	2	{f, c, a, b, m}	3	{f, b}	4	{c, p, b}	5	{f, c, a, m, p}				
TID	itemset																
1	{f, c, a, m, p}																
2	{f, c, a, b, m}																
3	{f, b}																
4	{c, p, b}																
5	{f, c, a, m, p}																
	Construct the FP tree. Show the trees separately after reading each transaction. (Min. Support=1)	6	L2	3	1												
c)	Write Apriori Algorithm for frequent itemset generation. Illustrate the same with the help of an example.	10	L2	3	1												
6. a)	How can the strength of an Association rule be measured? Also Mention the significance of these measures.	4	L2	3	1												
b)	Draw an itemset lattice structure for $I=\{a,b,c,d,e\}$. Show the application of Support based pruning on the same. Also brief on Anti - monotonicity property.	6	L2	3	1												
c)	Explain the process of FP tree construction with an example.	10	L2	3	1												

P.T.O.

Unit – IV

7. a) State the Hunt's Algorithm for inducing decision trees. Explain the same by taking the example of loan repayment prediction.
 b) State the k-nearest neighbor classification algorithm. Explain the same with an example. Also mention the characteristics of nearest neighbor classifiers.

10 L2 4 1

10 L2 4 1

8. a) What are following with respect to classification?

- i. Training set
- ii. Test set
- iii. Accuracy
- iv. Error rate

- b) Write a short note on following:

- i. Confusion matrix
- ii. Mutually exclusive rules
- iii. Nearest neighbours

4 L2 4 1

- c) Explain the algorithm used to extract rules directly from data.

6 L2 4 1
10 L2 4 1

Unit – V

9. a) Explain following methods of defining proximity between clusters:

- i. Single link
- ii. Complete link
- iii. Group average

- b) Distinguish following types of clustering:

- i. Hierarchical and partitional clustering
- ii. Complete and partial clustering
- iii. Exclusive and overlapping clusters

6 L2 5 1

- c) Explain various classification-oriented measures of cluster validity.

6 L2 5 1
8 L2 5 1

10. a) Explain different types of clusters with suitable examples.

- b) Write the pseudocode for following clustering algorithms:

- i. Basic Agglomerative Hierarchical Clustering Algorithm
- ii. Basic K-means Algorithm

10 L2 5 1

10 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
